

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

충북대학교 AI Cross-lab tech seminar

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ORIGINAL VOCABULARY

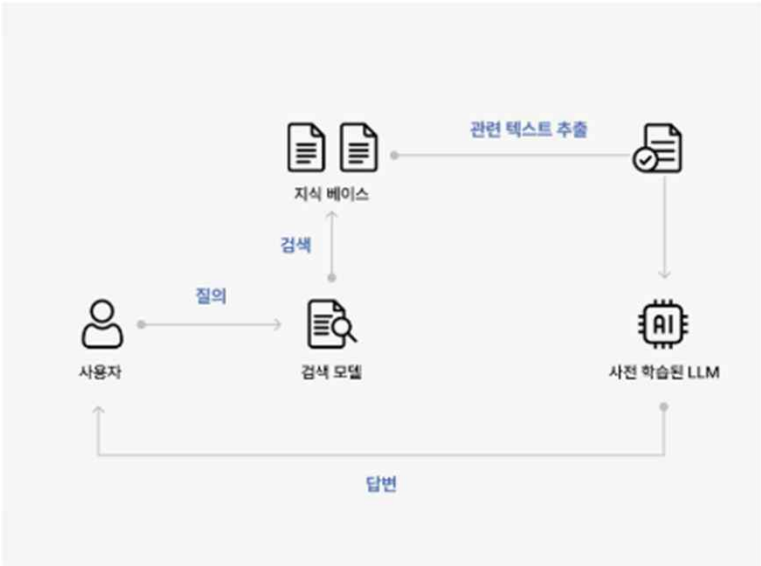
- 전체 어휘 수: 50,265
- 추가된 토큰: 0개

입력 문장	토큰화 결과 (Tokens)	토큰 수
thou shalt not covet	["th", "ou", "Gshalt", "Gnot", "Gove", "t"]	6
hath not a Jew eyes	["h", "ath", "Gnot", "G", "Jew", "Geyes"]	6
wherefore art thou Romeo	["where", "fore", "Gart", "G", "thou", "G Romeo"]	5

EXTENDED VOCABULARY

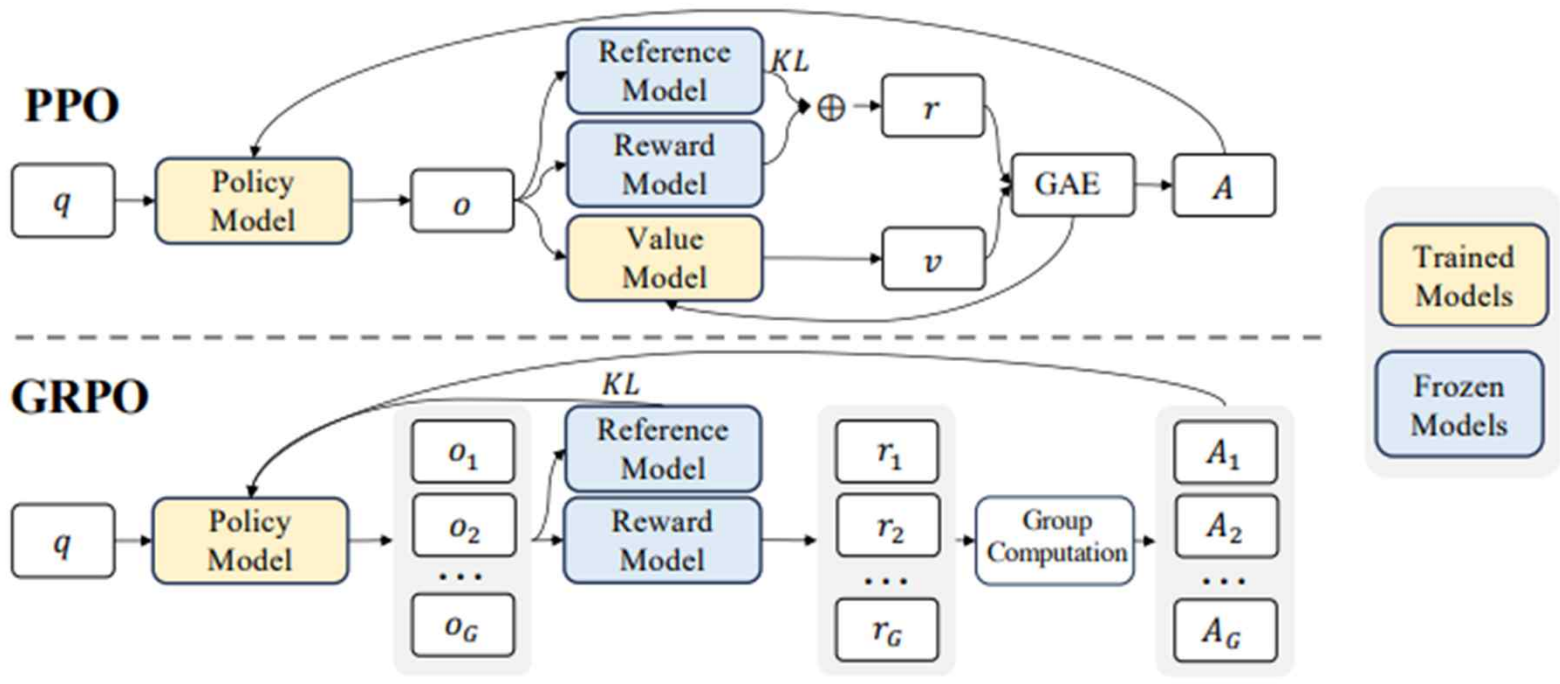
- 전체 어휘 수: 59,128
 - 추가된 토큰: 8,863개
- 새로 추가된 토큰 샘플: antime, aculate, DAMAG, Gcomely, Ghangings, Gpore, throat, Gpharaoh, Gheedless, GDavy

입력 문장	토큰화 결과 (Tokens)	토큰 수
thou shalt not covet	["thou", "Gshalt", "Gnot", "G", "cove", "t"]	6
hath not a Jew eyes	["hath", "Gnot", "Ga", "GJew", "Geyes"]	5
wherefore art thou Romeo	["where", "fore", "Gart", "G", "thou", "G", "Rome", "a"]	8



- **논문 제목:** DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning
- **저자 및 출판 정보**
 - 저자: DeepSeek-AI (주저자: Dayi Guo, Dejian Yang, Haowei Kaya 등 200여 명)
 - 출판/발표: 2025년 1월 22일 (arXiv 공개 및 *SCIENCE CHINA Technological Sciences* 게재)
- 예시: Mark D. Robin et al., Stanford univ. (1저자 소속), *학술대회이름* 2025
(Mark D. Robin et al., Stanford univ. (1저자 소속), *저널 이름*, 연도)
- **Contributions**
 - (1) Pure RL의 가능성 증명 (DeepSeek-R1-Zero)
 - (2) GRPO 알고리즘 도입
 - (3) Multi-stage Training Pipeline 구축
 - (4) 작은 모델들도 높은 추론 능력을 가질 수 있음을 증명

- Reasoning
 - Chain of Thought (CoT)
 - Self-Correction
 - Logical Rules
- Pre-training
 - Substantial computational resources
- CoT prompting
 - Dependence on human annotated reasoning trace
 - Capped by the human- provided exemplars



■ DeepSeek-R1-Zero

Group Relative Policy Optimization

Reward Design

$$Reward_{rule} = Reward_{acc} + Reward_{format}$$

$$J_{GRPO}(\theta) = E[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta D_{KL}(\pi_{\theta} || \pi_{ref}) \right)$$

$$D_{KL}(\pi_{\theta} || \pi_{ref}) = \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1$$

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}$$

■ DeepSeek-R1-Zero

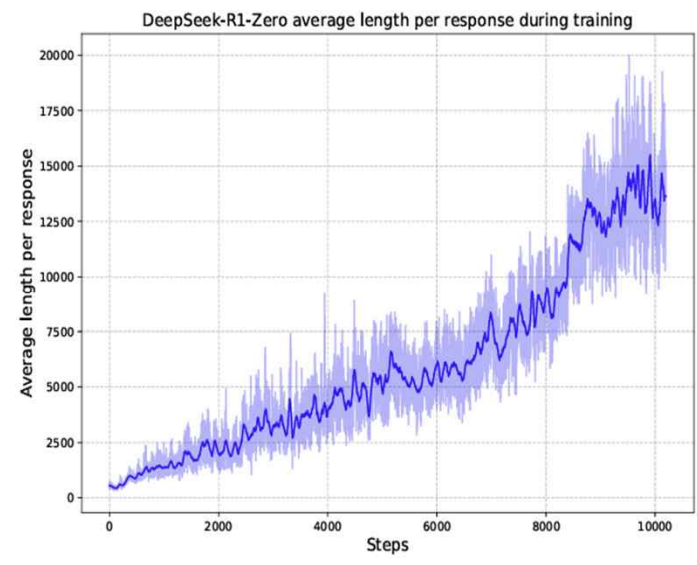
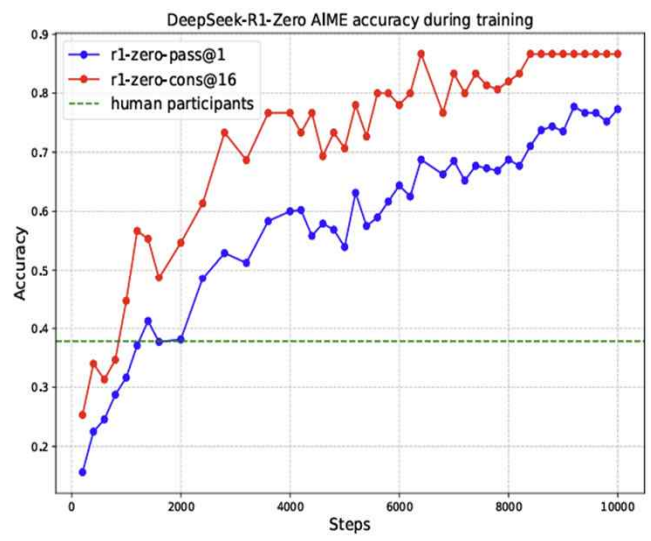


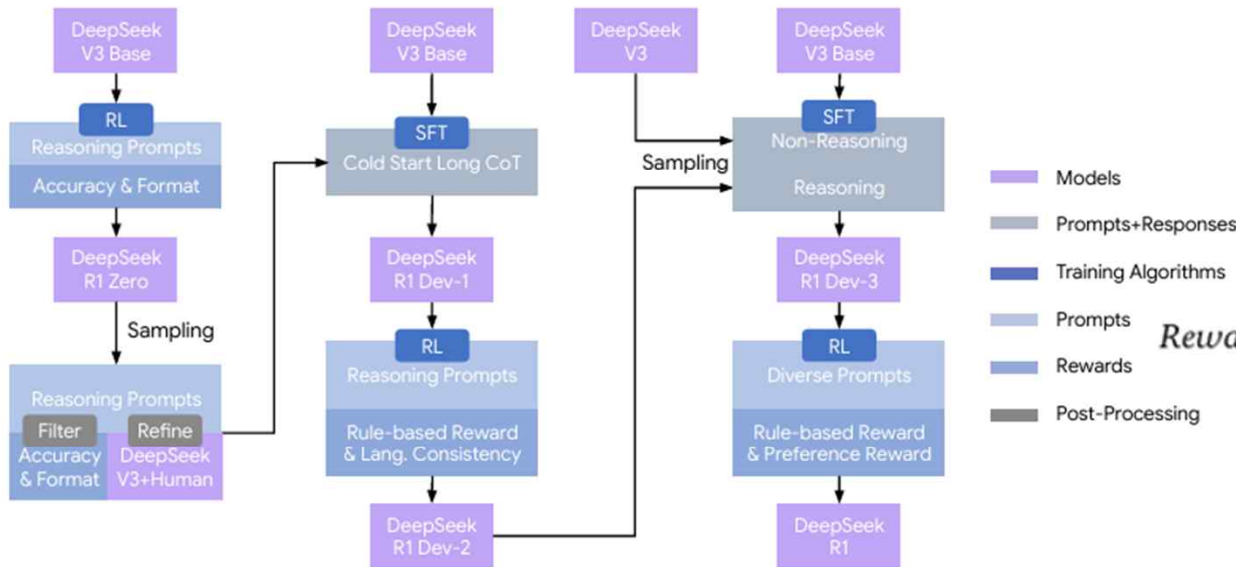
Table 2 | An interesting “aha moment” of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to
Response: <think>
To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...
$(\sqrt{a - \sqrt{a + x}})^2 = x^2 \implies a - \sqrt{a + x} = x^2$.
Rearrange to isolate the inner square root term:
$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$
...
Wait, wait. Wait. That's an aha moment I can flag here.
Let's reevaluate this step-by-step to identify if the correct sum can be ...
We started with the equation:
$\sqrt{a - \sqrt{a + x}} = x$
First, let's square both sides:
$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$
Next, I could square both sides again, treating the equation: ...
...

05 논문리뷰: Experiments

| 개인 연구 분야 | 요약 | 배경 | Method | Experiments | Limitations & Personal opinions |

■ DeepSeek-R1



$$Reward_{helpful} = RM_{helpful}(Response_A, Response_B)$$

$$Reward_{safety} = RM_{safety}(Response)$$

$$Reward_{language} = \frac{Num(Words_{target})}{Num(Words)}$$

$$Reward = Reward_{reasoning} + Reward_{general} + Reward_{language}$$

Figure 2 | The multi-stage pipeline of DeepSeek-R1. A detailed background on DeepSeek-V3 Base and DeepSeek-V3 is provided in Supplementary A.1. The models DeepSeek-R1 Dev1, Dev2, and Dev3 represent intermediate checkpoints within this pipeline.

05 논문리뷰: Experiments

	Benchmark (Metric)	R1-Zero	R1-Dev1	R1-Dev2	R1-Dev3	R1
English	MMLU (EM)	88.8	89.1	91.2	91.0	90.8
	MMLU-Redux (EM)	85.6	90.0	93.0	93.1	92.9
	MMLU-Pro (EM)	68.9	74.1	83.8	83.1	84.0
	DROP (3-shot F1)	89.1	89.8	91.1	88.7	92.2
	IF-Eval (Prompt Strict)	46.6	71.7	72.0	78.1	83.3
	GPQA Diamond (Pass@1)	75.8	66.1	70.7	71.2	71.5
	SimpleQA (Correct)	30.3	17.8	28.2	24.9	30.1
	FRAMES (Acc.)	82.3	78.5	81.8	81.9	82.5
	AlpacaEval2.0 (LC-winrate)	24.7	50.1	55.8	62.1	87.6
	ArenaHard (GPT-4-1106)	53.6	77.0	73.2	75.6	92.3
Code	LiveCodeBench (Pass@1-COT)	50.0	57.5	63.5	64.6	65.9
	Codeforces (Percentile)	80.4	84.5	90.5	92.1	96.3
	Codeforces (Rating)	1444	1534	1687	1746	2029
	SWE Verified (Resolved)	43.2	39.6	44.6	45.6	49.2
	Aider-Polyglot (Acc.)	12.2	6.7	25.6	44.8	53.3
Math	AIME 2024 (Pass@1)	77.9	59.0	74.0	78.1	79.8
	MATH-500 (Pass@1)	95.9	94.2	95.9	95.4	97.3
	CNMO 2024 (Pass@1)	88.1	58.0	73.9	77.3	78.8
Chinese	CLUEWSC (EM)	93.1	92.8	92.6	91.6	92.8
	C-Eval (EM)	92.8	85.7	91.9	86.4	91.8
	C-SimpleQA (Correct)	66.4	58.8	64.2	66.9	63.7

Table 8 | Comparison between DeepSeek-R1 and other representative models. Numbers in bold denote the performance is statistically significant (t-test with $p < 0.01$).

	Benchmark (Metric)	Claude-3.5-Sonnet-1022	GPT-4o-0513	DeepSeek-V3	OpenAI-o1-mini	OpenAI-o1-1217	DeepSeek-R1
English	Architecture	-	-	MoE	-	-	MoE
	# Activated Params	-	-	37B	-	-	37B
	# Total Params	-	-	671B	-	-	671B
	MMLU (EM)	88.3	87.2	88.5	85.2	91.8	90.8
	MMLU-Redux (EM)	88.9	88.0	89.1	86.7	-	92.9
	MMLU-Pro (EM)	78.0	72.6	75.9	80.3	-	84.0
	DROP (3-shot F1)	88.3	83.7	91.6	83.9	90.2	92.2
	IF-Eval (Prompt Strict)	86.5	84.3	86.1	84.8	-	83.3
	GPQA Diamond (Pass@1)	65.0	49.9	59.1	60.0	75.7	71.5
	SimpleQA (Correct)	28.4	38.2	24.9	7.0	47.0	30.1
Code	FRAMES (Acc.)	72.5	80.5	73.3	76.9	-	82.5
	AlpacaEval2.0 (LC-winrate)	52.0	51.1	70.0	57.8	-	87.6
	ArenaHard (GPT-4-1106)	85.2	80.4	85.5	92.0	-	92.3
	LiveCodeBench (Pass@1-COT)	38.9	32.9	36.2	53.8	63.4	65.9
	Codeforces (Percentile)	20.3	23.6	58.7	93.4	96.6	96.3
Math	Codeforces (Rating)	717	759	1134	1820	2061	2029
	SWE Verified (Resolved)	50.8	38.8	42.0	41.6	48.9	49.2
	Aider-Polyglot (Acc.)	45.3	16.0	49.6	32.9	61.7	53.3
Chinese	AIME 2024 (Pass@1)	16.0	9.3	39.2	63.6	79.2	79.8
	MATH-500 (Pass@1)	78.3	74.6	90.2	90.0	96.4	97.3
	CNMO 2024 (Pass@1)	13.1	10.8	43.2	67.6	-	78.8
Chinese	CLUEWSC (EM)	85.4	87.9	90.9	89.9	-	92.8
	C-Eval (EM)	76.7	76.0	86.5	68.9	-	91.8
	C-SimpleQA (Correct)	55.4	58.7	68.0	40.3	-	63.7

06 논문리뷰: Limitations & Personal opinions

| 개인 연구 분야 | 요약 | 배경 | Method | Experiments | Limitations & Personal opinions |

Limitations

- Structure Output and tool use
- Token efficiency
- Language mixing
- Prompt Engineering
- Software Engineering Tasks
-
- Reward Hacking

06 논문리뷰: Limitations & Personal opinions

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Personal opinions

- Rule-based Reward Dependency
- Inference Cost

Thank you for attention

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Q&A time

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